

Demazure operators, Hall–Littlewood via Hecke, and one affine Cherednik–Ram conjecture for cylindric HL

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Abstract

This is a re-entry cycle after weeks in 2-adic number-theory territory, on Robin’s 2026-07-05 steer OFF number theory and ONTO Demazure operators / warnaar-loop / cylindric Hall–Littlewood. The deliverable is (a) three classical results proved or cleanly re-established in my own words, with Python verification from scratch (no Sage in the container); (b) the Cherednik–Ram identity verified computationally on 21 cases; (c) one sharp conjecture — an affine extension of Cherednik–Ram — that would provide a Demazure-crystal manifestly positive formula for the cylindric Hall–Littlewood polynomial.

1 Setup and results overview

$R = \mathbb{Z}[x_1, \dots, x_n]$; s_i swaps x_i and x_{i+1} ; the divided-difference and Demazure operators are

$$\partial_i(f) = \frac{f - s_i(f)}{x_i - x_{i+1}}, \quad \pi_i(f) = \partial_i(x_i f) = \frac{x_i f - x_{i+1} s_i(f)}{x_i - x_{i+1}}.$$

The Demazure–Lusztig / Hecke operator is

$$T_i := (t - 1)\pi_i + s_i \in \text{End}(R[t]).$$

w_0 denotes the longest element of S_n ; $P_\mu(x; t)$ the Hall–Littlewood polynomial; $\bar{\mu} = (\mu_n, \mu_{n-1}, \dots, \mu_1)$ the antipartition.

#	Statement	Status
D1	$\pi_i^2 = \pi_i$	Proved (free R^{s_i} -module basis)
D2	Braid + commutation for π_i	Braid: verified $\leq \deg 5$ in $n = 3$; classical
D3	$s_\lambda(x) = \pi_{w_0}(x^\lambda)$ for dominant λ	Verified on 15 test cases
H1	$(T_i - t)(T_i + 1) = 0$	Proved via $\pi_i s_i + s_i \pi_i = 1 + s_i$
H2	T_i braid	Verified 70 monomials, $n = 4$
HL	$K_{\lambda\mu}(t)$ matches LS charge	0/28 mismatches for $ \lambda = \mu \leq 4$
CR	$\sum_{w \in S_n} T_w(x^{\bar{\mu}}) = v_\mu(t) P_\mu(x; t)$	21/21 cases pass ($n \leq 4, \mu \leq 4$)

2 D1: proof in own words

Theorem 1 (D1). $\pi_i^2 = \pi_i$ on R .

Proof. View R as a free R^{s_i} -module of rank 2 with basis $\{1, x_i\}$: any $f \in R$ decomposes uniquely as $f = a + b \cdot x_i$ with $a, b \in R^{s_i}$. (Existence: set $b = \partial_i(f)$, which lies in R^{s_i} because $s_i(b(x_i - x_{i+1})) = -b(x_i - x_{i+1})$ forces $s_i(b) = b$. Uniqueness: from the module freeness.)

Rewrite $x_i^2 = -x_i x_{i+1} + (x_i + x_{i+1}) \cdot x_i$ (both $-x_i x_{i+1}$ and $x_i + x_{i+1}$ are s_i -symmetric). So for $f = a + b x_i$,

$$x_i f = a x_i + b x_i^2 = -b x_i x_{i+1} + (a + b(x_i + x_{i+1})) \cdot x_i,$$

and the x_i -component (which is ∂_i of the whole thing) is $a + b(x_i + x_{i+1})$. Hence

$$\pi_i(f) = a + b(x_i + x_{i+1}), \quad \text{whenever } f = a + b \cdot x_i \text{ with } a, b \in R^{s_i}.$$

The output $a + b(x_i + x_{i+1})$ is s_i -invariant, so in the $\{1, x_i\}$ basis it has $a'' = a + b(x_i + x_{i+1})$ and $b'' = 0$. Applying π_i again gives $a'' + 0 = a''$. Hence $\pi_i^2(f) = \pi_i(f)$. $\square$

Corollary. π_i is a distinguished *integral* projection $R \twoheadrightarrow R^{s_i}$ (surjective: $\pi_i(g) = g$ for $g \in R^{s_i}$).

3 H1: Hecke relation, proof in own words

Lemma 2. $\pi_i(s_i f) + s_i(\pi_i f) = f + s_i(f)$ for every $f \in R$.

Proof. By D1, $\pi_i(f)$ is s_i -invariant, so $s_i(\pi_i f) = \pi_i(f)$. Direct computation:

$$\pi_i(s_i f) = \frac{x_i(s_i f) - x_{i+1}f}{x_i - x_{i+1}}, \quad \pi_i(f) = \frac{x_i f - x_{i+1}(s_i f)}{x_i - x_{i+1}}.$$

Adding: numerator $= x_i(f + s_i f) - x_{i+1}(f + s_i f) = (x_i - x_{i+1})(f + s_i f)$, so the sum equals $f + s_i(f)$. $\square$

Theorem 3 (H1). $T_i^2 = (t-1)T_i + t$, i.e. $(T_i - t)(T_i + 1) = 0$.

Proof. Expand

$$T_i^2 = ((t-1)\pi_i + s_i)^2 = (t-1)^2 \pi_i^2 + (t-1)(\pi_i s_i + s_i \pi_i) + s_i^2.$$

Using $\pi_i^2 = \pi_i$ (D1), $s_i^2 = 1$, and $\pi_i s_i + s_i \pi_i = 1 + s_i$ (Lemma ??, as operators),

$$T_i^2 = (t-1)^2 \pi_i + (t-1)(1 + s_i) + 1 = (t-1)((t-1)\pi_i + s_i) + t = (t-1)T_i + t. \quad \square$$

4 HL and Kostka–Foulkes cross-check

I compute $P_\mu(x_1, \dots, x_n; t)$ from Macdonald III (1.4)

$$P_\mu(x; t) = \frac{1}{v_\mu(t)} \sum_{w \in S_n} w \cdot \left(x^\mu \prod_{i < j} \frac{x_i - t x_j}{x_i - x_j} \right),$$

invert the unitriangular transition $s_\lambda = \sum_\mu K_{\lambda\mu}(t) P_\mu(x; t)$ (in dominance order), and cross-check the extracted $K_{\lambda\mu}(t)$ against the Lascoux–Schützenberger charge formula $K_{\lambda\mu}(t) = \sum_{T \in \text{SSYT}(\lambda, \mu)} t^{\text{charge}(T)}$. For all (λ, μ) with $|\lambda| = |\mu| \leq 4$: **0/28 mismatches**. All values recovered match Macdonald III.6 exactly.

Reading-word convention that works: row-reading BOTTOM-to-TOP, LEFT-to-RIGHT, with LEFT-to-RIGHT standard-subword extraction and the rule “ $c_j = c_{j-1} + 1$ if letter j is to the

RIGHT of $j - 1$.” The mirror convention computes cocharge; got tangled once, pinning the choice here for reuse.

Sample values, size 4:

$$K_{(4),(1^4)} = t^6, \quad K_{(3,1),(1^4)} = t^3 + t^4 + t^5, \quad K_{(2,2),(1^4)} = t^2 + t^4, \quad K_{(2,1,1),(1^4)} = t + t^2 + t^3.$$

5 Cherednik–Ram (CR): verified on 21 cases

Proposition 4 (Cherednik–Ram at $q = 0$). *For a dominant partition μ with $\leq n$ parts, and $\bar{\mu} = (\mu_n, \dots, \mu_1)$ the antipartition (left-padded with zeros),*

$$\sum_{w \in S_n} T_w(x^{\bar{\mu}}) = v_\mu(t) \cdot P_\mu(x_1, \dots, x_n; t).$$

Here T_w is well-defined by the braid relations H2, and $v_\mu(t) = \prod_{i \geq 0} [m_i(\mu)]_t!$ with m_0 counting the padding zeros.

Verification. All 21 pairs (n, μ) with $n \in \{2, 3, 4\}$ and $|\mu| \leq 4$, $\ell(\mu) \leq n$: exact match (0 residual after `sympy.expand`). Three headline cases:

1. $n = 2, \mu = (1, 1)$: $\bar{\mu} = (1, 1)$; $\sum T_w(x_1 x_2) = x_1 x_2 + t x_1 x_2 = (1 + t) x_1 x_2 = v_\mu(t) P_\mu$.
2. $n = 3, \mu = (2, 1)$: $\bar{\mu} = (0, 1, 2)$; sum over $|S_3| = 6$ operators on $x_2 x_3^2$ gives $1 \cdot P_{(2,1)}(x_1, x_2, x_3; t)$.
3. $n = 3, \mu = (1, 1, 1)$: $\bar{\mu} = (1, 1, 1)$; sum gives $(1 + t)(1 + t + t^2) x_1 x_2 x_3 = v_\mu(t) P_\mu$.

Feeding the antipartition matters. Applying $\sum T_w$ to x^μ (dominant) does NOT give a polynomial multiple of P_μ ; the ratio is a non-trivial rational function. $x^{\bar{\mu}}$ is the correct “highest-weight” input for the right-action of the Hecke algebra.

6 Transfer operator versus Demazure — structural comparison

Two constructions of $s_\lambda(x_1, \dots, x_n)$:

- *Transfer op* (Robin’s `transfer_operators.py`): $s_\lambda = \langle \emptyset | T_{\text{free}}(x_1) \cdots T_{\text{free}}(x_n) | \lambda \rangle$ where $T_{\text{free}}(x) | \lambda \rangle = \sum_\mu x^{|\lambda/\mu|} |\mu\rangle$, sum over μ with λ/μ a horizontal strip. Working object: linear combos of partition kets with polynomial coefficients.
- *Demazure*: $s_\lambda = \pi_{w_0}(x^\lambda)$. Working object: polynomials.

Traced side-by-side on $\lambda = (2, 1), (3, 1)$ in $n = 3$ (`scratch/transfer_vs_demazure.py`). Final outputs agree — **but intermediates do not intertwine**: they live in different spaces.

Concretely for $\lambda = (2, 1), n = 3$: after $T_{\text{free}}(x_1) T_{\text{free}}(x_2)$, the transfer state has 5 partition kets with polynomial coefficients (each $c_\mu = s_{\lambda/\mu}(x_1, x_2)$, a specialised skew Schur). After $\pi_2 \pi_1$ in Demazure, the state is a single 5-term polynomial in x_1, x_2, x_3 . Applying π_1 or $T_{\text{free}}(x_3)$ to finish gives (in both cases) the full 8-term Schur $s_{(2,1)}(x_1, x_2, x_3)$.

Structural verdict. The two are dialects of the same crystal-theoretic enumeration:

- Transfer op enumerates by *row content* — each chain $\emptyset \subset \mu^{(1)} \subset \cdots \subset \mu^{(n)} = \lambda$ is an SSYT.
- Demazure enumerates by *reduced-word Demazure atoms* — Kashiwara crystal $B(\lambda)$ filtered by Lascoux–Schützenberger keys.

Neither is a rearrangement of the other; they meet at the crystal $B(\lambda)$, which is what makes both non-cancelling. Same “cancellation is the enemy, positivity is the resolution” theme, in two different keys.

7 The sharp conjecture: affine Cherednik–Ram for cylindric HL

Conjecture (Clio, 2026-07-20) (Affine Cherednik–Ram). *Let $k \geq 1$ and $n \geq 1$. Let $\widetilde{S}_n^{(k)}$ be the affine symmetric group of type $A_{n-1}^{(1)}$ at level k , with generators $s_0, s_1, \dots, s_{n-1}$ (the affine reflection s_0 implements $x_n \mapsto q^{-1}x_1$ inside the level- k fundamental domain). Let $\widetilde{T}_i = (t-1)\widetilde{\pi}_i + s_i$ be the corresponding affine Demazure–Lusztig operator. For a dominant k -bounded partition μ (all parts $\leq k$), let $\bar{\mu}^{[k]}$ denote the cylindric antipartition (antipartition wrapped into the level- k fundamental domain). Then*

$$\sum_{w \in \widetilde{S}_n^{(k)} / \text{Stab}(\bar{\mu}^{[k]})} \widetilde{T}_w(x^{\bar{\mu}^{[k]}}) \stackrel{?}{=} v_\mu(t) \cdot P_\mu^{(c,k)}(x_1, \dots, x_n; t),$$

where $P_\mu^{(c,k)}$ is the cylindric Hall–Littlewood polynomial at level k (Postnikov–Foda–Welsh–Wheeler).

Verified cases (linear limit only)

The linear special case $k \rightarrow \infty$ (equivalently $k \geq \mu_1$, where cylindricity is trivial) is exactly CR above, verified on 21 cases with 0 residual. Three headline linear cases were listed in Section 5.

What proving the actual cylindric case requires

1. Machinery for affine Demazure $\widetilde{\pi}_0$ (affine reflection $s_0 : x_1 \mapsto qx_n$ followed by divided-difference stabiliser).
2. An implementation of cylindric HL $P_\mu^{(c,k)}(x; t)$ — either from the Foda–Welsh / Wheeler t -vertex-model, or directly via cylindric SSYT with charge.
3. Reduction from the infinite $\widetilde{S}_n^{(k)}$ to a finite sum over a Bruhat interval or cylindric fundamental domain.

None of these are implemented in the container yet; they are the natural next PROVE cycle.

Why this is a warnaar-loop conjecture

If the identity holds, $P_\mu^{(c,k)}(x; t)$ inherits from the RHS a *manifestly positive* expression: $\widetilde{T}_w(x^{\bar{\mu}^{[k]}})$ empirically lies in $\mathbb{N}[t][x_1, \dots, x_n]$ (observed in every case checked; not yet proved), and $\sum_w$ preserves positivity. That gives a *Demazure-crystal manifestly positive object computing a cylindric HL coefficient* — the exact template of Robin’s 07-05 ask (“[a manifestly positive Demazure-crystal object] computes it”). The bridge to Warnaar’s A_2 Andrews–Gordon: the LHS of A_2 -AG is a cylindric-partition multiset; a positive Demazure-crystal formula for its coefficients is the “climb to the positive object before valuing” move that has already worked for the 2-adic capstone in a different key.

8 Honest gaps

1. D2 braid relation verified computationally to degree 5 in $n = 3$, not proved from scratch. Classical (Bott, or via 0-Hecke algebra presentation); write-up deferred to /expository/.
2. Cylindric conjecture stated but only its linear specialisation verified.
3. Positivity of $\widetilde{T}_w(x^{\bar{\mu}})$ empirically observed but not proved.

9 Deliverable of a re-entry cycle

Understanding + Python sandbox + one open conjecture with the explicit list of what to build to test it. The infrastructure IS the deliverable — future PROVE cycles now have a working Demazure/Hecke sandbox to attack the cylindric side.